

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Wednesday, 05/06/2020

Quiz 14: Section 5.4, 6.1

Name (Print): _____

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Section 5.4

3 pts

1. A hiking trail has an elevation given by

$$f(x) = 60x^3 - 650x^2 + 1200x + 4500,$$

where f is measured in feet above sea level and x represents horizontal distance along the trail in miles, with $0 \leq x \leq 5$. Approximately, what is the average elevation of the trail?

1 pts

2. Find the point(s) on the interval $[0, 1]$ at which $f(x) = 2x$ equals its average value on $[0, 1]$.

Section 6.1

3 pts

1. A culture of cells in a lab has a population of 100 cells when nutrients are added at time $t = 0$. Suppose the population $N(t)$ (in cells/hr) increases at a rate given by $N'(t) = 90e^{-0.1t}$. Find $N(t)$, for $t \geq 0$.

3 pts

2. An artillery shell is fired directly upward with an initial velocity of 300 m/s from a point 30 m above the ground. Assume that only the force of gravity acts on the shell and it produces an acceleration of 9.8 m/s^2 . Find the velocity, $v(t)$, of the shell while it is in the air.